BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 082 · Lesson 082 · Quicksort: Partition Around a Pivot

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is merge sort's combine step?

Q2. What is a pivot in quicksort?

Q3. Is a partitioned array necessarily fully sorted?

Q4. Swap values at indexes0 and2 in [4,1,7]. Give the resulting array.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

quicksort — Recursive partition-based sorting.

pivot — Value partitioning subarray.

partition — Place smaller/larger values around pivot.

worst case — Maximum growth for adverse input.

New learning

Observe — Partitioning reorders values around a chosen pivot so values to its left are no larger and those right are larger, for this partition rule.

Apply — The pivot's final index is fixed after partition, but neither side is necessarily fully sorted; quicksort continues with the two subarrays.

Evaluate — Average and best-case time O(n log n), worst $O(n^2)$ with poor pivot choices; recursion consumes stack space depending on partition balance.

Worked example

Partition [6,2,5,1,3] using last pivot3; give partition state.

1. Method and result — Lomuto partition moves 2 and 1 left, pivot3 to index2 → [2,1,3,6,5].

We Do · Practise with guidance

Partition [8,3,6,2,5] using last pivot5.

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Smaller 3,2 move left; pivot5 at index2 → [3,2,5,8,6].
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Method and result

Smaller 3,2 move left; pivot5 at index2 → [3,2,5,8,6].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Trace Lomuto pivot-last partition of [5,2,7,1,4] and give final pivot index. [6]

Question 2

After quicksort partition gives [2,1,4,5,7] with pivot4 at index2, state required recursive subranges and eventual sorted result. [6]

Question 3

Explain how pivot choice impacts quicksort worst-case time and recursive stack depth. [6]

Exit Ticket · Answer before checking

Once a pivot reaches its correct final index, both partitions are also completely sorted.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Merge two individually sorted runs.

Q2. Selected value used to partition an array.

Q3. No; subarrays still need sorting.

Q4. [7,1,4].

You Do pages · Independent questions

Q1. Moves2 then1 [2], partition [2,1,4,5,7] [3], pivot index2 [1].

Q2. Left indexes0–1 [2], right3–4 [2], sorted result [1,2,4,5,7] [2].

Q3. Unbalanced partitions n−1 and0 [2], O(n^2) time [2], up to O(n) recursion depth [2]; balanced partitions tend O(log n) depth.

Exit Ticket · Model answer

False. Both sides may still need sorting.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: